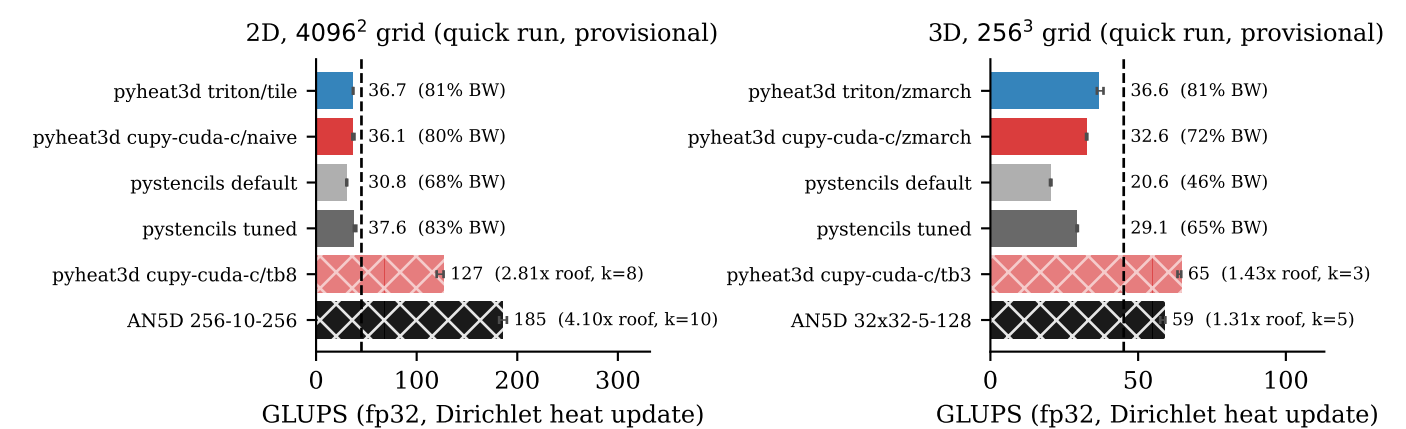

Same GPU, same session, same update. Colours: pyheat3d backends; grey/black: established codes. Dashed: single-step fp32 roofline (copy BW / 8 B). 'xx': temporal blocking (k steps per DRAM pass). Whiskers: min-max of 5 repeats.